

CSE4502 (Operating Systems Lab): 1A

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1 Part A: File Guard (Conditionals)

1.1 Problem Statement

Develop a Bash script named `file_guard.sh` that verifies whether a specified file exists and determine whether the provided file name corresponds to a regular file, a directory, or other.

1.2 Logic

1. Take a filename as a command-line argument.
2. Use `if [ -e $filename ]` to check existence.
3. If it exists, check if it is a directory (`-d`) or a file (`-f`).

1.3 Hints

- Use `$1` to access the first argument passed to the script.

1.4 Example Output

Case 1: File exists

```
$ ./file_guard.sh notes.txt
'notes.txt' is a regular file.
```

Case 2: Directory exists

```
$ ./file_guard.sh Documents
'Documents' is a directory.
```

Case 3: File does not exist

```
$ ./file_guard.sh unknown.txt
Error: 'unknown.txt' does not exist.
```

2 Part B : Thread-Safe Histogram

2.1 Problem Statement

A large dataset is processed by multiple threads. They update a shared frequency array (histogram) of size K .

```
histogram[10] = {0, 0, ...}
```

2.2 Rules

- Input is an array of n integers which have values k ranging from $0 \leq k < K$. k is an integer.
- Assume you have $n = 9$. If you create 2 threads then thread $t1$ will calculate the histogram for indices $0 - 4$ and thread $t2$ will calculate the histogram for indices $5 - 8$, both done in a histogram array local to their threads. Then both the threads concurrently update the global histogram using the locally generated histograms.

2.3 Inputs & Outputs

- **Input:** First line of input contains n <space> K .
- **Output:** Histogram[i]: count for i in $0..(K - 1)$.

2.4 Constraints

- Use **one mutex**.

2.5 Invariant to Preserve

$$\sum \text{histogram}[i] = n$$